

Sentinel: An Agentic System for DeFi Risk Monitoring and Decision Recommendation Based on a Graph Time-Series Foundation Model

Anonymous ACL submission

Abstract

General-purpose LLM agents asked to recommend financial-risk interventions can *hallucinate severity*: handed evidence of elevated risk, they over-read it, recommending high-stakes actions whose every cited number is traceable to the input yet whose targets are never actually stressed. We design and characterise a *deployment-ready* LLM-agent pipeline for decentralised-finance supervision that measures and contains this failure mode: a graph time-series forecaster produces predicted exposure networks, deterministic monitors and stress scenarios convert them into a typed evidence bundle, and the LLM emits supervisory tickets with rationales. Two safety gates—a data-health gate and a confidence gate—determine when intervention-level tickets are allowed; every emitted ticket carries the evidence bundle and rationale that produced it. We release an evaluation harness that scores such pipelines on six axes, anchored to a *regulator-aligned absolute-loss* ground truth that replaces the fractional-change metric used in prior systemic-risk evaluations, and additionally measures LLM-as-judge explanation quality and a false-intervention rate (FIR). With eight reference implementations and five years of weekly DeFi exposure graphs, the full LLM-agent stack raises ticket F1 from 0.0076 to 0.0233 (+207%) at a visible false-intervention cost (FIR $\approx$ 0.44 under an Opus 4.7 decision LLM vs 0 for rules). A tier-above Opus 4.8 judge and a cross-family panel (Gemini 2.5 Pro, GPT-5.5) all rank the full stack top on explanation quality. Replacing the decision LLM with a tier-below Sonnet 4.6 yields a Pareto improvement over Opus 4.7 on every LLM-bearing axis (F1 0.0288, FIR 0.000, judge 2.65) at $\sim 5\times$ lower inference cost. We characterise the resulting system as a recall–explanation Pareto option for supervisory settings, not a uniform replacement for conservative rules, and release ticket-level audit logs from the historical back-test.

1 Introduction

In May 2022 the Terra/Luna collapse erased over \$40 billion in 48 hours, triggering cascading liquidations through lending protocols, DEXs, and yield aggregators that held UST-denominated assets (Vidal-Tomás et al., 2023). Five months later FTX wiped out another \$8 billion. These crises were not isolated: in decentralised finance (DeFi), credit exposures form a dense, time-varying network of tokenised claims that no single participant fully observes, and shocks travel through it within hours.

A natural response is to build an *agent*: an LLM-driven pipeline that ingests on-chain measurements, surfaces early-warning signals, and recommends risk-mitigating actions with auditable rationale. Our reference implementations (§5) show that the obvious designs fail in two ways. Reading raw on-chain snapshots, the LLM is conservative but weak: poorly targeted tickets (F1 0.0183), and the lowest-scored rationales under every judge we test. Fed richer *forecasted* evidence, it targets better but *hallucinates severity*: every cited number is traceable to the evidence bundle (grounding 1.000), yet 44.8% of its intervention-level tickets point at protocols that are never actually stressed. In a supervisory setting the second failure is far costlier than missed detections, because every intervention is observable and politically expensive.

To measure such a failure mode honestly we first need a ground truth that matches how supervisors actually prioritise. Prior systemic-risk evaluations rank protocols by *fractional* weight change (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025), which over-weights tiny protocols of low systemic relevance and is not what a supervisor ranks by. **We define a regulator-aligned *absolute-loss* ground truth** (§4) and use it to score, for the first time, an integrated forecast+monitor+scenario+LLM+gate pipeline

against what a supervisor would actually prioritise. The characterisation produces the first joint measurement of forecast quality, monitor lead time, and ticket quality on the same regulator-aligned target, and the first quantification of the false-intervention cost for an LLM agent operating in this setting—a cost that finance-specific LLM-agent work (Yao et al., 2026; Mao et al., 2025; Luo et al., 2025; Jeon and Lee, 2026) has not measured, since none of those systems is scored against a regulator-aligned ground truth.

We close this gap with three contributions:

- **A deployment-ready LLM-agent supervisory pipeline** (Sentinel; §3). A graph time-series foundation model (Forecast-FM) produces predicted exposure networks; five deterministic monitors and five stress scenarios convert them into a typed evidence bundle; an LLM emits supervisory tickets with rationale; and two safety gates (data-health, confidence) determine when intervention-level tickets are allowed.
- **An evaluation harness** (Decision-Bench; §4). Six axes spanning forecast quality, streaming anomaly detection, predictive uncertainty, what-if scenario fidelity, supervisory ticket quality, and data-quality robustness, with a regulator-aligned *absolute-loss* ground truth (§4) and eight reference method implementations including persistence, EvolveGCN (Pareja et al., 2020), rule engines, snapshot LLMs, and Forecast-FM-coupled LLM variants.
- **An operating-point characterisation** (§5). On the frozen 2025 split of Exposure-Corpus, the full Forecast-FM+LLM+gate stack raises ticket F1 from 0.0076 to 0.0233 (+207%). A tier-above Claude Opus 4.8 judge (Zheng et al., 2023) and a cross-family panel (Gemini 2.5 Pro, GPT-5.5) all rank the gated stack m7 highest on explanation quality; under the Opus 4.7 decision LLM the false-intervention rate rises to ≈ 0.44 . A tier-below Sonnet 4.6 decision LLM yields a Pareto improvement on both m6 and m7: F1 rises further to 0.0288, FIR collapses to 0.000, and judge quality reaches 2.65, at $\sim 5\times$ lower inference cost—the deployable headline operating point.

Honest scope and track fit. We have run the pipeline as a historical backtest over 283 weekly snapshots of real DefiLlama-sourced exposure data (2020–2025), not as a live production cron at a supervisory authority. The contribution is therefore

best read as *methods for deployed systems (evaluation, ethics, robustness)* and *insights based on real-world datasets*—two of the Industry Track’s explicit categories—rather than as a deployment case study. The released ticket-level audit logs come from the backtest, letting operators recompute the operating point under their own data conditions before adopting the system.

2 Related Work

Bench positioning and ground-truth definition.

LLM-agent benchmarks (HELM (Liang et al., 2022), SWE-bench (Jimenez et al., 2024), Agent-Bench (Liu et al., 2024)) score open-ended reasoning, software repair, and generic agent behaviour; temporal-graph benchmarks (TGB (Huang et al., 2023), OGB (Hu et al., 2020)) score structural prediction quality. Neither scores whether an LLM agent’s supervisory decisions match what a regulator would actually prioritise. The closest prior systemic-risk evaluations (Bertomeu et al., 2024; Gonon et al., 2025; Li et al., 2025) rank protocols by *fractional* weight change, which disproportionately surfaces tiny protocols of low systemic relevance. Decision-Bench replaces this with an *absolute-loss* ground truth (§4) so that every downstream axis is scored against what supervisors actually rank by.

LLM agents in finance and DeFi. General agent patterns such as ReAct (Yao et al., 2023) combine reasoning with tool use; FinGPT (Yang et al., 2023) adapts language models to financial data. Within DeFi, agentic systems have so far targeted transaction auditing (Yao et al., 2026), intent mining (Mao et al., 2025), smart-contract verification (Hu et al., 2026; Kong et al., 2026), price manipulation (Liu et al., 2025), and explanation for graph anomalies (Watson et al., 2025); portfolio-oriented agents combine graph encoders with LLMs (Luo et al., 2025; Jeon and Lee, 2026). These systems consume raw transactions or token text, not structured forecasted evidence, and none reports a false-intervention rate against a regulator-aligned ground truth—a gap our characterisation (§5) closes.

Forecast-grounded LLM agents in other domains. Pairing a domain forecaster with an LLM decision layer is an emerging deployment pattern. In macroeconomics, ChatGPT-augmented PMI nowcasting (de Bondt and Sun, 2025) and LLM-driven macroeconomic forecasting (Carriero

et al., 2025) both feed structured numeric inputs into an LLM that produces interpretable narratives; a BIS primer surveys the wider pattern (Kwon et al., 2024). Time-series foundation models such as Chronos (Ansari et al., 2024), Lag-Llama (Rasul et al., 2024), and TimesFM (Das et al., 2024) make the forecaster reusable across tasks; tabular foundation models extend the same idea to heterogeneous structured data (Hollmann et al., 2025; Ere-meev et al., 2025). None of these forecast→LLM pipelines, to our knowledge, has been scored against a regulator-aligned ground truth for high-stakes financial-network supervision.

3 Pipeline

Sentinel is a four-layer pipeline (Figure 1). Layer 1 forecasts the near-future exposure graph with Forecast-FM; Layer 2 converts the forecast into a typed evidence bundle of monitor alerts and stress scenarios; Layer 3 hands the bundle to an LLM that emits supervisory tickets with rationale; Layer 4 gates intervention-level actions on a data-health and a confidence check. The pipeline runs at weekly cadence on a single RTX 4090; a decision epoch takes under a minute of GPU time and well under a dollar of LLM API spend (§4).

3.1 Forecaster interface

At each decision epoch t the agent observes a credit-exposure snapshot $G_t = (V, E_t, W_t)$ (an inter-protocol token-mediated exposure graph; see §A) and node covariates X_t . It queries Forecast-FM for a forecast distribution $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$ at horizons $\mathcal{H} = \{1, 4, 8, 12\}$ weeks, constructs an expected-weight predicted graph $\hat{G}_{t+h}$ by thresholding edge-existence probabilities at $\pi_{\min} = 0.2$, and draws $M = 50$ Monte Carlo samples for uncertainty. Full predicted-graph construction equations are in §A.

3.2 Structured evidence

Layer 2 turns $\hat{G}_{t+h}$ and the MC samples into a typed evidence bundle. Five *monitor functionals* score concentration, fragility, and instability of the predicted graph: top-1 PageRank (Brin and Page, 1998), HHI concentration (Rhoades, 1993), network density, PageRank Gini, and degree Gini. An alert fires when a monitor metric departs from its $L = 42$ week rolling baseline by more than $z = 2$ standard deviations; each alert is attached to (i) an attribution top- K ($K = 5$) of contributing edges

by marginal contribution, and (ii) an uncertainty-aware confidence score $C_t(a) \in [0, 1]$ that decays with low data-health, wide MC dispersion, and longer horizons.

In parallel, a *scenario engine* applies five standardised stress shocks—single protocol failure, bridge cluster failure, stablecoin de-peg, sector lending shock, correlated top-10 stress—to both $\hat{G}_{t+h}$ and each MC sample, and aggregates the worst-half losses as $\text{CVaR}_{0.5}(s, h)$. Full monitor and scenario definitions are in §B.

3.3 LLM decision layer

Layer 3 receives the evidence bundle as a typed JSON payload—alerts with attribution top- K , scenario CVaR table, MC dispersion, and the scalar data-health score DH_t —and is asked to emit a ranked list of supervisory tickets. Each ticket carries (a) a severity level drawn from a fixed four-action playbook (Monitor, Investigate, Recommend-Reduce, and Contingency), (b) a target set of protocols, and (c) a free-form rationale that cites specific evidence fields by name. Self-consistency is probed with three repeated calls at temperature 0 (Jaccard overlap on targets and severities).

3.4 Safety gates

Two gates determine the feasibility of intervention-level actions. The *data-health gate* $DH_t = \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) < \tau_{\text{data}} = 0.7$ triggers safe mode, in which only Monitor and Investigate tickets are emitted, regardless of LLM recommendation. The *confidence gate* $\bar{C}_t < \tau_{\text{conf}} = 0.6$ blocks intervention-level tickets even when data is healthy. Every emitted ticket carries the full evidence bundle, rationale, and gate states; the audit log is the unit of release (§6).

The end-to-end algorithm is in §B (Algorithm 1).

4 Evaluation Harness

Why a new harness. Scoring an LLM-agent supervisory pipeline requires evidence on six independent axes that no existing benchmark covers together: how good is the forecast (b1_forecast), how early does the monitor turn (b2_warning), is the predictive uncertainty calibrated (b3_calibration), do scenario losses match reality (b4_stress), do the emitted tickets match what a regulator would prioritise (b5_decision), and what happens under degraded

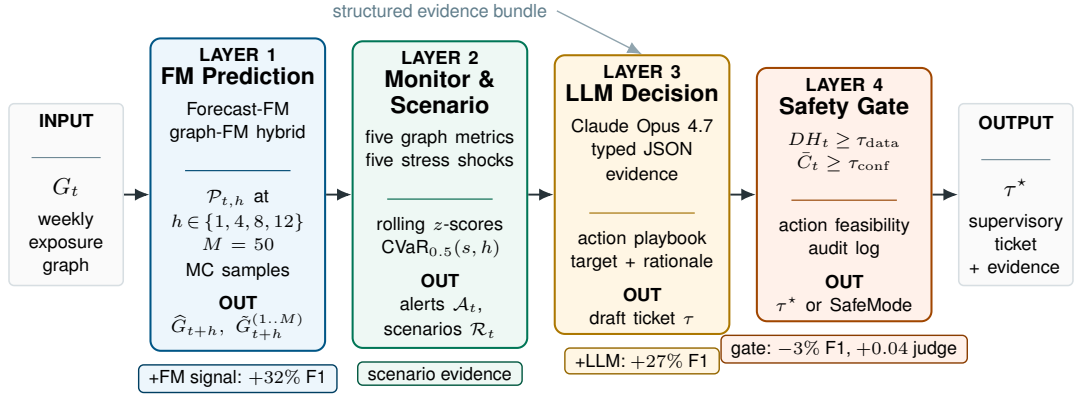


Figure 1: Sentinel: a four-layer pipeline. The agent ingests the weekly exposure graph G_t , runs Forecast-FM prediction (Layer 1), monitor and scenario engines (Layer 2), an LLM decision layer (Layer 3), and a safety gate (Layer 4) that emits either the supervisory ticket τ^* or a safe-mode notice. Bottom annotations report the per-layer F1 / judge lift from §5.

data (b6_robustness). No existing benchmark covers these axes together (§2). Decision-Bench unifies the six axes on a shared dataset and reference-method set so that ablations on the forecaster, monitor, scenario, and decision layers can be reported independently.

Dataset and split. We use the Exposure-Corpus dataset of weekly directed weighted exposure graphs (4,300 protocols, 24,300 tokens, 43.7M exposure entries, 283 weekly snapshots from 2020-03 to 2025-08 (Wu et al., 2025)). We adopt a strict no-leakage temporal split: 2020-03–2024-06 train (~222 weeks), 2024-07–2024-12 validation (~26 weeks, used for conformal calibration and threshold tuning), 2025-01–2025-08 test (~33 weeks, frozen leaderboard). All Forecast-FM fine-tuning and model selection likewise use pre-2025 data only (§A). For b2_warning we additionally run a historical event study around Terra/Luna (2022-05-09), FTX (2022-11-07), and SVB/USDC (2023-03-10); the event study is reported separately from the 2025 predictor-specific leaderboard.

Regulator-aligned absolute-loss ground truth. Prior systemic-risk evaluations rank protocols by *fractional* weight change, which silently up-weights tiny protocols of low systemic relevance. Decision-Bench ranks instead by *absolute* loss: with $w_t(v)$ the total weight incident on v at t , the stressed set at horizon h is $\mathcal{S}_t^h = \text{top}_\pi\{v: w_t(v) > 0, \Delta_t^h(v) > 0\}$, where $\Delta_t^h(v) = w_t(v) - w_{t+h}(v)$. We fix $\pi = 0.05$ and the ground-truth horizon $h = 4$ weeks throughout (yielding $|\mathcal{S}_t^h| \approx 283$ protocols per week); the last four test weeks lack a 4-week-ahead snapshot, so decision axes

score $n = 29$ weeks. The same ground truth anchors b2_warning, b5_decision, and the LLM-decision evaluation pipeline.

Eight reference implementations. The harness bundles eight reference methods spanning the four ablation cuts of the four-layer framework: heuristic streaming monitor (h1), persistence+rules (m1), snapshot LLM (m2), EvolveGCN (Pareja et al., 2020) (m3), Forecast-FM-only (m4), Forecast-FM+rules (m5), Forecast-FM+LLM (m6), and Forecast-FM+LLM+gate (m7). Each method exposes the same predict_graph or decide_actions interface so adding a ninth method is a single-file change. The full method specification and per-axis schema are in §C–§D.

Reproducibility. A complete leaderboard run finishes in under four GPU-hours on a single RTX 4090 plus approximately \$10 of LLM API spend, making the harness accessible without industrial compute. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files; every reported number can be reproduced bit-exactly from the released artefacts.

5 Empirical Results

We use Decision-Bench to ask three questions. Does Forecast-FM provide forecast signal that persistence does not (b1, b3, b6)? Does routing the LLM through structured evidence improve supervisory tickets once precision, recall, explanation quality, and false interventions are scored jointly (b4, b5)? Which components are load-bearing, and which are reserves under degradation (ablations)?

Forecast-FM contributes trend, calibration, and robustness; persistence still wins on static point error. On static error metrics, persistence is a deceptively strong baseline: at $h = 4$ it beats the FM on PageRank MAE (3.4×10^{-5} vs 4.5×10^{-5}) and edges it on Spearman rank correlation (0.570 vs 0.558). The FM contributes signal where persistence is structurally pinned: trend consistency 0.525–0.632 across horizons (persistence: 0); b3_calibration PI coverage 0.913 against a 0.90 target and ECE 0.013; b6_robustness mean relative degradation 0.113 vs 0.148 for persistence (24% less). The full per-horizon and per-regime audits are in §E.

Headline decision-quality result. Table 1 consolidates the central b5_decision comparison. The persistence+rules baseline (m1) achieves the highest single-axis precision (0.720) and a false-intervention rate of 0 by virtue of raising few tickets. Against this baseline, the full Forecast-FM+LLM+gate stack (m7) lifts ticket F1 from 0.0076 to 0.0233 (+207%; $p < 10^{-4}$). Against the comparable pure-LLM baseline that consumes raw snapshots without the forecaster (m2, the only other LLM-bearing row), m7 lifts LLM-as-judge explanation quality from 2.24 to 2.45 on a 1–5 scale (+9.4%) under a tier-above Claude Opus 4.8 judge (Zheng et al., 2023); the lift is significant under GPT-5.5 ($p < 0.001$) but not Opus 4.8 ($p = 0.23$; §F)—directional rather than confirmed. The cost surfaces sharply once m2 is in the table: adding the FM signal to the LLM (m2→m6) raises the false-intervention rate from 0.000 to 0.448 ($p < 10^{-4}$). The safety gate (m6→m7) reduces FIR only marginally (0.448 → 0.437), and its effects on F1 and judge score are within noise (§F). Both LLM variants reach grounding 1.000, so the failure mode is not citation hallucination but *over-reading the forecaster’s evidence as warrant for high-severity action*.¹

Judge robustness across model families. We re-judge m2/m6/m7 with a cross-family panel (Table 2). All three frontier judges rank m7 highest, and its mean score varies by at most 0.45 across the three. The largest disagreement is that Gemini 2.5 Pro scores m2 slightly above m6 (2.69 vs 2.55), still keeping m7 on top; Gemini is also $\sim 3\times$ noisier per

¹m3_evolvegc and m4_fm_only are predictor-only and do not emit tickets (forecast quality in §E.1); EvolveGCN trails persistence downstream, consistent with its ~ 220 weekly-snapshot training budget.

week (std 1.3–1.5 vs 0.3–0.6), so the inversion sits within judge noise. Cross-family agreement on the top system closes the judge-family confound.

A tier-below, cheaper decision LLM dominates the headline operating point. We re-run the decision layer of m6 and m7 on the *same* FM-derived prompts with two non-Opus 4.7 decision LLMs—Claude Sonnet 4.6 (within-family, tier-below, $\sim 5\times$ cheaper) and Google Gemini 2.5 Pro (cross-family)—and re-score with the same Opus 4.8 judge (Table 3). On both architectures Sonnet 4.6 is a Pareto improvement over Opus 4.7: F1 rises (+15%/+24%; $p < 0.001$ on m7), FIR collapses from 0.437–0.448 to 0.000, and judge quality is directionally higher (+0.11/+0.20; n.s.). The cross-family Gemini 2.5 Pro decision is more conservative still (lower F1, FIR = 0.000), indicating that over-intervention is a property of the decision model rather than of LLM decision layers per se. The operating point a practitioner would now deploy is m7 with Sonnet 4.6 (F1 0.0288, FIR 0.000, judge 2.65).

Layer-wise contribution. Holding the decision policy fixed at LLM and adding the Forecast-FM signal lifts ticket F1 by 32% (0.0183 → 0.0241; $p < 0.001$); holding the predictor fixed at Forecast-FM and swapping rules for the LLM lifts F1 by another 27% (0.0190 → 0.0241). The two layers therefore make additive, non-substitutable contributions (Figure 2 in §E). Adding the safety gate trades 3% F1 for judge-quality changes within noise; its value is operational—every intervention-level ticket clears two pre-defined gates with auditable logs.

Matched-budget view. F1 conflates target quality with volume: m6/m7 emit 6.3–6.5 targets per week vs 4.9 for m2 and 2.1 for m1. Truncating every method to its top- k targets per week ($k \in \{1, 2, 3\}$; §F.3) separates the two effects: the LLM variants still roughly double the rules baseline’s precision@ k ($p \leq 0.0055$), but the FM-fed and raw-snapshot LLM variants become statistically indistinguishable. The FM’s contribution is thus not a sharper top of the ranking but support for a *larger* target set at undiminished precision; under Opus 4.7 the expanded tail is where over-intervention concentrates, while Sonnet 4.6 expands it without escalating severity.

Ablation: load-bearing vs reserve components. Under clean 2025 data the confidence gate is load-

ID	Pred.	Decision	Prec.↑	Recall↑	F1↑	Judge↑	FIR↓	Ground.↑	Stabil.↑
m1	persist.	rules	0.720	0.004	0.0076	—	0.000	—	0.514
m2	—	LLM (no FM)	0.575	0.009	0.0183	2.24	0.000	1.000	0.532
m5	Forecast-FM	rules	0.600	0.010	0.0190	—	0.000	—	0.257
m6	Forecast-FM	LLM	0.570	0.012	0.0241	2.41	0.448	1.000	0.488
m7	Forecast-FM	LLM + safety gate	0.580	0.012	0.0233	2.45	0.437	1.000	0.435

Table 1: Headline b5_decision comparison on the frozen 2025 test split ($n = 29$). Rows are read as a multi-objective trade-off rather than a single leaderboard: m1 is the conservative high-precision baseline; m2 is the pure-LLM baseline (no forecaster); m6/m7 are the FM-grounded LLM variants. Judge scores (1–5 scale) are from a Claude Opus 4.8 judge, strictly tier-above the Claude Opus 4.7 decision LLM; cross-family agreement is in Table 2.

Method	Opus 4.8	Gemini 2.5 Pro	GPT-5.5	Haiku 4.5 [†]
m2	2.24	2.69	1.90	2.03
m6	2.41	2.55	2.45	2.48
m7	2.45	2.90	2.45	2.69

Table 2: Cross-family judge panel on the same Opus 4.7 decisions ($n = 29$). Gemini and GPT use reasoning = {effort : minimal} so the thinking budget matches the Anthropic baseline. [†] Haiku 4.5 is a tier-below weak-judge reference.

Method	Decision LLM	F1↑	FIR↓	Judge↑
m6	Opus 4.7 (main)	0.0241	0.448	2.41
m6	Sonnet 4.6	0.0276	0.000	2.52
m6	Gemini 2.5 Pro	0.0129	0.000	2.21
m7	Opus 4.7 (main)	0.0233	0.437	2.45
m7	Sonnet 4.6	0.0288	0.000	2.65
m7	Gemini 2.5 Pro	0.0139	0.000	2.38

Table 3: Cross-decision-LLM panel on the same FM-derived prompts, judge fixed at Claude Opus 4.8 ($n = 29$). Sonnet 4.6 dominates Opus 4.7 on both architectures. The bold row (m7 × Sonnet 4.6) beats every row in Table 1 on every LLM-bearing axis at $\sim 5\times$ lower cost.

bearing: removing it raises FIR from 0.000 to 0.429. The scenario engine is load-bearing for target extraction: skipping it collapses ticket precision from 0.600 to 0.000. The data-health gate and multi-horizon diversity are dormant on clean data but activate as reserves under degradation. Under severe feature/edge masking (80–98%) the data-health gate suppresses up to 60% of interventions that would otherwise produce false alarms (FIR range [0.27, 0.60] when disabled vs 0.000 when triggered); multi-horizon monitoring generates $\sim 4\times$ more alerts in the three crisis windows at unchanged precision. Per-regime ablation audits are in §E.6.

Early-warning timeline (historical event study). For the three pre-test crisis windows the shared

weighted-degree monitor (h1) achieves median lead times of 4–5 weeks with precision 1.000 at SVB/USDC across all alert budgets $K \in \{5, 10, 20\}$; per-budget details in §E.4. This is historical monitor evidence, not an FM-specific test-split result.

6 Deployment Lessons

Three lessons should transfer to other forecast→LLM pipelines for high-stakes structured-evidence decisions.

L1. Grounding $\neq$ correctness. Every LLM variant we test achieves grounding 1.000: every cited number in every ticket traces to a field of the input evidence bundle. The FM-grounded variants still pay FIR = 0.448/0.437 (ungated/gated) under Opus 4.7. The failure is not invented evidence but *over-reading* of weak-but-present evidence as justification for high-severity action. Grounding says the agent did not hallucinate citations; only a downstream intervention-realism metric (FIR against an absolute-loss ground truth) says whether it is safe to act on.

L2. Audit logs are a design artefact, not an output. Every ticket carries the forecast distribution, monitor scores, scenario CVaRs, LLM rationale, gate states, and seeds that produced it. We release these logs because our operating point is not what every operator will need: a more conservative supervisor can retune τ_{data} or τ_{conf} and rerun the decision policy post-hoc, without re-invoking the FM or the LLM.

L3. Safety gates are reserves, not always-on filters. On clean 2025 data the data-health gate is dormant; a leaderboard without a degradation track cannot see it. Under masking it suppresses up to 60% of FIR-eligible tickets at the cost of suspending interventions—so evaluate gates on dedicated stress regimes, not only clean splits.

Limitations

Backtest, not field deployment. All results come from a historical backtest over 283 weekly snapshots of Exposure-Corpus (2020–2025). We have not run Sentinel as a live weekly cron at a supervisory authority. The ticket-level audit logs we release come from the backtest. Field deployment lessons—operator behaviour, end-to-end latency under real data feeds, regulator-in-the-loop calibration—are out of scope for this paper and remain the most pressing future direction.

Single domain. Decision-Bench evaluates a single on-chain risk surface (DeFi inter-protocol credit exposure). Generalisation to NFT lending, perpetual derivatives, or cross-chain bridge networks is untested and will require domain-specific recalibration of the conformal split and the stressed-set percentile π .

Weekly resolution. The pipeline operates at weekly granularity, but several DeFi shocks unfold within hours—Terra/Luna erased \$40B in under 48 hours. A sub-hourly production deployment would require denser snapshot ingestion and re-evaluation of conformal calibration on the resulting denser time series. The 4–5 week lead times we report should be read as “how early the weekly monitor turns”, not as end-to-end alert latency.

Judge thinking-budget is held minimal. The cross-family judge panel (Claude Opus 4.8, Google Gemini 2.5 Pro, OpenAI GPT-5.5) is run with reasoning effort set to *minimal* so the comparison against the non-reasoning Anthropic baseline is a like-for-like comparison of intrinsic model capability rather than thinking budget. Default reasoning mode on Gemini 2.5 Pro and GPT-5.5 would change the reported scores; we treat the reasoning-on configuration as a distinct evaluation surface and leave it for a follow-up study. Practitioners deploying the judge at default reasoning settings should re-calibrate the score thresholds reported here. Human-judge calibration with practising supervisors remains the most pressing methodological extension.

Offline decision evaluation. `b5_decision` scores tickets against a regulator-aligned absolute-loss ground truth, which captures supervisory targeting priority but not the downstream effect of supervisory action—whether the recommended risk reduction would actually have averted conta-

gion. A controlled live evaluation with a regulator in the loop, which Decision-Bench approximates but cannot replace, remains the primary direction we plan to extend the harness in.

Ethics Statement

Data provenance. Exposure-Corpus is built from public on-chain measurements aggregated via DefiLlama (DefiLlama, 2026). No private user data, wallet-level personally-identifying information, or off-chain identity data is used. Protocol-level aggregates are the unit of analysis throughout.

Decision risk. Sentinel emits supervisory tickets, not automated trades or automated interventions. The high-severity action types (Recommend-Reduce, Contingency) are gated on a data-health score and a confidence score, and the pipeline emits a safe-mode notice when either gate fails. We characterise the system as decision support for a human supervisor, not as an autonomous risk agent. The measured false-intervention rate ($\text{FIR} \approx 0.44$ for the Forecast-FM+LLM variant) is reported in the headline results table so that downstream operators can weigh the trade-off against their own risk tolerance before deployment.

LLM use disclosure. The decision-layer LLM (`m2`, `m6`, `m7`) is Claude Opus 4.7 in the main configuration; Claude Sonnet 4.6 and Google Gemini 2.5 Pro are reported as decision-layer ablations. The LLM-as-judge for the explanation-quality axis is a three-model cross-family panel: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family). All judges are run with reasoning effort set to *minimal* so the comparison is controlled for thinking budget. Claude Haiku 4.5 from a prior run is reported in the appendix as a weak-judge baseline. Every model is accessed through the OpenRouter unified API at temperature 0.

Release. We will release the harness, the eight reference implementations, and the ticket-level audit logs under a permissive open licence. Model checkpoints will be released subject to dataset-licence compatibility checks.

References

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Olek-

- sandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiesner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. 2024. Chronos: Learning the language of time series. *Preprint*, arXiv:2403.07815.
- Jeremy Bertomeu, Xiumin Martin, and Ibrahima Sall. 2024. Measuring defi risk. *Finance Research Letters*, 63:105321.
- Sergey Brin and Lawrence Page. 1998. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1):107–117. Proceedings of the Seventh International World Wide Web Conference.
- Andrea Carriero, Davide Pettenuzzo, and Shubhanshu Shekhar. 2025. Macroeconomic forecasting with large language models. *Preprint*, arXiv:2407.00890.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. 2024. A decoder-only foundation model for time-series forecasting. *Preprint*, arXiv:2310.10688. Accepted at ICML 2024; model commonly referenced as TimesFM.
- Gabe J. de Bondt and Yiqiao Sun. 2025. Enhancing GDP nowcasts with ChatGPT: a novel application of PMI news releases. Working Paper Series 3063, European Central Bank, Frankfurt am Main, Germany. ECB Working Paper Series No. 3063; released 2025-06-30.
- DefiLlama. 2026. DefiLlama: DeFi TVL aggregator and analytics. Data used: 2025; accessed: 2026-02-02.
- Dmitry Ereemeev, Gleb Bazhenov, Oleg Platonov, Artem Babenko, and Liudmila Prokhorenkova. 2025. Turning tabular foundation models into graph foundation models. *Preprint*, arXiv:2508.20906.
- Lukas Gonon, Thilo Meyer-Brandis, and Niklas Weber. 2025. Computing systemic risk measures with graph neural networks. *Preprint*, arXiv:2410.07222.
- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. 2025. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326. Published online 8 Jan 2025.
- Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. 2020. Open graph benchmark: Datasets for machine learning on graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Xiaohui Hu, Wun Yu Chan, Yuejie Shi, Qumeng Sun, Wei-Cheng Wang, Chiachih Wu, Haoyu Wang, and Ningyu He. 2026. An effective and cost-efficient agentic framework for ethereum smart contract auditing. *Preprint*, arXiv:2601.17833.
- Shenyang Huang, Farimah Poursafaei, Jacob Danovitch, Matthias Fey, Weihua Hu, Emanuele Rossi, Jure Leskovec, Michael Bronstein, Guillaume Rabusseau, and Reihaneh Rabbany. 2023. Temporal graph benchmark for machine learning on temporal graphs. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Joohyoung Jeon and Hongchul Lee. 2026. Can blindfolded LLMs still trade? An anonymization-first framework for portfolio optimization. *Preprint*, arXiv:2603.17692.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. 2024. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations (ICLR)*.
- Ziqiao Kong, Wanxu Xia, Chong Wang, Yi Lu, Pan Li, Shaohua Li, Zong Cao, and Yang Liu. 2026. Knowdit: Agentic smart contract vulnerability detection with auditing knowledge summarization. *Preprint*, arXiv:2603.26270.
- Byeungchun Kwon, Taejin Park, Fernando Perez-Cruz, and Phurichai Rungcharoenkitkul. 2024. Large language models: a primer for economists. *BIS Quarterly Review*.
- Xurui Li, Xin Shan, Wenhao Yin, and Haijiao Wang. 2025. Heterogeneous graph pre-training based model for secure and efficient prediction of default risk propagation among bond issuers. *Preprint*, arXiv:2501.03268.
- Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, and 1 others. 2022. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*.
- Lu Liu, Wuqi Zhang, Lili Wei, Hao Guan, Yongqiang Tian, Yepang Liu, and Shing-Chi Cheung. 2025. LLM-powered detection of price manipulation in DeFi. *Preprint*, arXiv:2510.21272.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2024. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations (ICLR)*.
- Yichen Luo, Yebo Feng, Jiahua Xu, Paolo Tasca, and Yang Liu. 2025. Llm-powered multi-agent system for automated crypto portfolio management. *arXiv preprint arXiv:2501.00826*.
- Qian’ang Mao, Yuxuan Zhang, Jiaman Chen, Wenjun Zhou, and Jiaqi Yan. 2025. Know your intent: An autonomous multi-perspective LLM agent framework for DeFi user transaction intent mining. *Preprint*, arXiv:2511.15456.

- Aldo Pareja, Giacomo Domeniconi, Jie Chen, Tengfei Ma, Toyotaro Suzumura, Hiroki Kanezashi, Tim Kaler, Tao Schardl, and Charles Leiserson. 2020. EvolveGCN: Evolving graph convolutional networks for dynamic graphs. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pages 5363–5370.
- Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, Mohammad Javad Darvishi Bayazi, George Adamopoulos, Roland Riachi, Nadhir Hassen, Marin Biloš, Sahil Garg, Anderson Schneider, Nicolas Chapados, Alexandre Drouin, Valentina Zantedeschi, Yuriy Nevmyvaka, and Irina Rish. 2024. Lag-llama: Towards foundation models for probabilistic time series forecasting. *Preprint*, arXiv:2310.08278.
- Stephen A. Rhoades. 1993. The herfindahl-hirschman index. *Federal Reserve Bulletin*, 79(3):188–189.
- David Vidal-Tomás, Antonio Briola, and Tomaso Aste. 2023. Ftx’s downfall and binance’s consolidation: The fragility of centralised digital finance. *Physica A: Statistical Mechanics and its Applications*, 625:129044.
- Adriana Watson, Grant Richards, and Daniel Schiff. 2025. Explain first, trust later: LLM-augmented explanations for graph-based crypto anomaly detection. *arXiv preprint arXiv:2506.14933*.
- Wenbin Wu, Kejiang Qian, Alexis Lui, Christopher Jack, Yue Wu, Peter McBurney, Fengxiang He, and Bryan Zhang. 2025. Dexposure: A dataset and benchmarks for inter-protocol credit exposure in decentralized financial networks. *Preprint*, arXiv:2511.22314.
- Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-source financial large language models. *Preprint*, arXiv:2306.06031.
- Duanyi Yao, Siddhartha Jagannath, Baltasar Aroso, Vyas Krishnan, and Ding Zhao. 2026. Auditable LLM arbiter for DeFi security: A hybrid graph-of-thoughts approach to intent-transaction alignment. In *NDSS 2026 LAST-X Workshop*.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhaghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS)*.

A Forecaster Interface and Notation

A.1 Credit exposure

Let $X_G(q)$ be the set of tokens issued by protocol q and $X_p(t)$ the set of tokens held by protocol p at time t . Protocol p has a credit exposure to protocol q at t if $X_p(t) \cap X_G(q) \neq \emptyset$ (Wu et al., 2025). This token-mediated dependence generalises counterparty credit risk from traditional finance to DeFi.

A.2 Network representation

Over an interval $\tau = t_2 - t_1$, exposures form a weighted directed graph $G_\tau = (P_\tau, E_\tau)$. Node weights are $w_\tau(p) = \sum_{\sigma \in X_p(t_1) \cap X_p(t_2)} v_{t_2, \sigma}$ where $v_{t_2, \sigma}$ is the USD value of token σ at t_2 ; edge weights aggregate non-negative token flows from p to q mapped to issuer q , as defined in Wu et al. (2025). We adopt the same construction without modification.

A.3 Forecast primitive and predicted graph

At each decision epoch t the agent queries $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$. From $\mathcal{P}_{t,h}$ we build a representative predicted graph via a thresholded expected-weight operator,

$$\hat{w}_{pq,t+h} := \Pr(e_{pq,t+h} = 1) \cdot \mathbb{E}[w_{pq,t+h} \mid e_{pq,t+h} = 1], \quad (1)$$

$$\hat{E}_{t+h} := \{(p, q) : \Pr(e_{pq,t+h} = 1) > \pi_{\min}\}, \quad (2)$$

with $\pi_{\min} = 0.2$ (tuned on validation). For uncertainty quantification we draw $M = 50$ Monte Carlo samples $\{\tilde{G}_{t+h}^{(m)}\}$ by Bernoulli edge sampling and Gaussian weight perturbation of $\mathcal{P}_{t,h}$.

A.4 Hybrid prediction strategy

In practice Forecast-FM keeps all edges from G_t , reweights them as $w'_{pq} = w_{pq} + \Pr(e_{pq,t+h} = 1) \cdot \hat{r}_{pq}$ where $\hat{r}_{pq}$ is the FM residual, and additionally adds new edges whose existence probability exceeds $\pi_{\min}$. This preserves network structure while letting the FM adjust weights and predict new connections.

A.5 Architecture and training

Forecast-FM is built on the GraphPFN graph-tabular foundation backbone (Hollmann et al., 2025; Ereemeev et al., 2025) with a LiMiX-16M tabular transformer encoder: 12 layers, 4 attention heads, hidden dimension 64. The same backbone serves three heads—edge existence (binary), edge weight (regression), and node-TVL change (regression)—trained jointly with a BCE+MSE composite loss. We fine-tune one checkpoint per horizon $h \in \{1, 4, 8, 12\}$ for 20 epochs with AdamW (learning rate 10^{-4} for heads, 10^{-5} for the backbone). Fine-tuning draws on Exposure-Corpus snapshots from March 2020 to January 2025: 104 training weeks, 12 validation weeks (used for early stopping), and 8 subsequent held-out weeks for internal evaluation. All data that influences weights or checkpoint selection precedes the frozen 2025 leaderboard split (§C). Full-graph in-context inference fits a single RTX 4090; one weekly forecast across all four horizons completes in well under a minute.

Held-out link-prediction quality is AUPRC = 0.978 / 0.973 / 0.967 / 0.967 and AUROC = 0.996 / 0.995 / 0.993 / 0.993 at $h = 1 / 4 / 8 / 12$ respectively; a frozen GraphPFN baseline (no fine-tuning) achieves AUPRC 0.936–0.940 at the same horizons, isolating the contribution of domain fine-tuning. These intrinsic forecaster numbers do not appear in the main text because the paper’s contribution is the operating point of the full pipeline, not the FM’s headline accuracy.

B Pipeline Mathematics

B.1 Data-health gate

The data-health score combines four deterministic checks of the incoming snapshot:

$$DH_t = \text{mean}(\text{fresh}_t, \text{miss}_t, \text{topo}_t, \text{disc}_t) \in [0, 1]. \quad (3)$$

Safe mode is triggered when $DH_t < \tau_{\text{data}} = 0.7$: monitor alerts are still emitted but intervention-type tickets are suppressed.

B.2 Monitor metrics

ID	Metric	Definition
N1	Systemic Importance	$\max_{p \in V} \text{PageRank}(\hat{G}_{t+h})_p$
N2	HHI Concentration	$\sum_p (d_p / \sum_q d_q)^2$, d_p weighted out-degree
N3	Network Density	$ E_{t+h} / V (V - 1)$
N4	PageRank Gini	$\text{Gini}(\{\text{PR}_p\}_{p \in V})$
N5	Degree Gini	$\text{Gini}(\{d_p\}_{p \in V})$

PageRank uses power iteration with damping 0.85 and 10 iterations. An alert fires when $|N_{t,h}^{(j)} - \mu_t^{(j)}| / \max(\sigma_t^{(j)}, \epsilon) > z$ with $z = 2$, $L = 42$ week rolling baseline, $\epsilon = 10^{-8}$. The b2_warning historical event study uses a shorter 26-week (≈ 6 -month) baseline window.

B.3 Uncertainty-aware alert confidence

With $\text{CV}_{t,h}^{(j)}$ the coefficient of variation of metric j across the M MC samples (clamped to $[0, 1]$),

$$C_t(a) := DH_t \cdot \frac{1}{1 + \text{CV}_{t,h}^{(j)}} \cdot \frac{1}{1 + \ln(1 + h)} \in [0, 1]. \quad (4)$$

$C_t(a)$ decreases monotonically with lower data health, wider predictive spread, and longer horizons.

B.4 Scenario engine

ID	Scenario	Target	Shock
S1	Single protocol fail	Highest-weight node	100%
S2	Bridge cluster fail	All bridge nodes	100%
S3	Stablecoin de-peg	All stablecoin nodes	50%
S4	Sector lending shock	All lending nodes	30%
S5	Correlated stress	Top-10 by weight	20%

Per-scenario loss $L_{t,h}(s) = (\sum_e w_e(\hat{G}_{t+h}) - \sum_e w_e(\hat{G}_{t+h}^s)) / \sum_e w_e(\hat{G}_{t+h})$; CVaR aggregation averages the worst $\lceil \lambda M \rceil$ MC-sample losses with $\lambda = 0.5$.

B.5 Decision feasibility and ticket scoring

An intervention-level action u is feasible iff

$$(\text{SAFE_MODE}_t = 0) \wedge (\bar{C}_t \geq \tau_{\text{conf}} = 0.6), \quad (5)$$

with $\bar{C}_t$ the mean alert confidence. Feasible tickets are scored as $\text{Score}_t(u) = w_u \cdot \bar{C}_t \cdot \text{Imp}_t$, where $w_u \in \{0.25, 0.50, 0.75, 1.00\}$ is the playbook severity weight and Imp_t is the mean CVaR across scenarios (default 1.0 when no scenarios are available).

B.6 End-to-end procedure

C Benchmark Details

C.1 Six-axis schema

ID	Capability tested	Primary metrics
b1_forecast	Temporal graph prediction	PageRank/HHI/Gini MAE, trend consistency, Spearman; per $h \in \{1, 4, 8, 12\}$
b2_warning	Streaming anomaly detection	Precision, recall, F1, lead time, alert stability; per event $\times$ budget
b3_calibration	Predictive uncertainty	PI coverage at 0.90 target, PI width, ECE, CRPS
b4_stress	What-if scenario fidelity	Loss MAE, distressed-count MAE, propagation depth MAE, target overlap@ k
b5_decision	Supervisory ticket quality	Precision, recall@ k , F1, target stability, judge score, FIR
b6_robustness	Data quality sensitivity	Per-benchmark metrics under 5 degradation regimes; relative degradation

The six axes are chosen to be (i) *non-overlapping*—a system can excel at b1 while failing b5; (ii) *decomposable along the four-layer framework*, so that ablations on the predictor layer (b1, b3, b6), monitor layer (b2), scenario layer (b4), and decision layer (b5) can be reported independently; and (iii) *individually publishable*, so that downstream work may extend a single axis without re-running the whole suite.

Algorithm 1 Sentinel: one decision epoch.

Require: (G_t, X_t) ; horizons $\mathcal{H}$; scenarios $\mathcal{S}_0$; thresholds $(\tau_{\text{data}}, \tau_{\text{conf}}, z, \pi_{\text{min}})$; MC count M
Ensure: Alerts $\mathcal{A}_t$; scenario summary $\mathcal{R}_t$; tickets $\mathcal{D}_t$
1: $DH_t \leftarrow \text{DataHealth}(G_t, X_t)$; $\text{SAFE_MODE}_t \leftarrow \mathbb{I}\{DH_t < \tau_{\text{data}}\}$
2: **for all** $h \in \mathcal{H}$ **do**
3: $\mathcal{P}_{t,h} \leftarrow \text{Forecast-FM.Forecast}(G_t, X_t, h)$
4: $\hat{G}_{t+h} \leftarrow \text{BuildPredGraph}(\mathcal{P}_{t,h}; \pi_{\text{min}})$
5: $\{\tilde{G}_{t+h}^{(m)}\} \leftarrow \text{MCSample}(\mathcal{P}_{t,h})$
6: Compute $\mathbf{N}_{t,h} = \Phi(\hat{G}_{t+h}, X_t)$, attach rolling stats, derive alerts $\mathcal{A}_{t,h}$ and confidences $C_t(\cdot)$
7: **for all** $s \in \mathcal{S}_0$ **do**
8: Compute $L_{t,h}(s), \tilde{L}_{t,h}^{(m)}(s), \text{CVaR}_{0.5}(s, h)$
9: **end for**
10: **end for**
11: $\mathcal{D}_t \leftarrow \text{PlaybookDecision}(\mathcal{A}_t, \mathcal{R}_t, DH_t, \text{SAFE_MODE}_t, \tau_{\text{conf}})$
12: **return** $(\mathcal{A}_t, \mathcal{R}_t, \mathcal{D}_t)$

C.2 Dataset split

Split	Period	Weeks	Usage
Train	2020-03–2024-06	~222	Forecaster training
Validation	2024-07–2024-12	~26	Conformal calibration, threshold tuning
Test	2025-01–2025-08	~33	Frozen leaderboard

C.3 Ground-truth equation

With $w_t(v) = \sum_{e \ni v} w_t(e)$ the total weight incident on v , the regulator-aligned stressed set at horizon h is

$$\Delta_t^h(v) = w_t(v) - w_{t+h}(v), \quad \mathcal{S}_t^h = \text{top}_\pi \{v : w_t(v) > 0, \Delta_t^h(v) > 0\}, \quad (6)$$

with $\pi = 0.05$ and ground-truth horizon $h = 4$ weeks. The final four test weeks lack the 4-week-ahead snapshot w_{t+4} and are not scored, leaving $n = 29$ evaluated weeks. This is the single ground-truth definition used by b2, b5, and the LLM-decision evaluation pipeline.

C.4 Reporting protocol

A submission produces one JSON file per {benchmark, method} pair following the harness schema, plus a single results.json aggregating the suite. The full leaderboard requires b1, b3, b5, b6; b2 and b4 are required only for systems that implement a monitor or scenario layer respectively. The harness fixes random seeds, snapshots library versions, and emits SHA-256 hashes of all checkpoint files.

D Reference Methods

ID	Predictor	Policy	Role
h1_weighted_degree	—	heuristic alert	Streaming anomaly baseline (b2)
m1_persistence_rules	$G_{t+h} = G_t$	rule engine	Decision baseline
m2_snapshot_llm	current snap	LLM	LLM-only ablation (no forecast)
m3_evolvegcnn	EvolveGCN	—	Learned-GNN baseline
m4_fm_only	Forecast-FM	—	FM-only ablation
m5_fm_rules	Forecast-FM	rule engine	FM + rules
m6_fm_llm	Forecast-FM	LLM	Full FM + LLM stack
m7_fm_llm_gated	Forecast-FM	LLM + safety gate	Safety-gated FM + LLM

The eight methods expose all four cuts of the four-layer ablation: adding the predictor layer (m1→m5), swapping the decision layer (m5→m6), adding the safety gate (m6→m7), and removing the predictor from the LLM stack (m2→m6). All methods conform to a shared predict_graph/decide_actions interface; adding a ninth method is a single-file change.

LLM model panel and decoding. All LLM-bearing methods (m2, m6, m7) are evaluated under three decision-layer configurations: Claude Opus 4.7 (main), Claude Sonnet 4.6 (within-family, tier-below ablation), and Google Gemini 2.5 Pro (cross-family ablation). Each runs at temperature 0 and

max_tokens = 4096; self-consistency is probed with three repeated calls per week and Jaccard overlap on emitted ticket targets and severities.

The LLM-as-judge for the explanation-quality axis is a three-model panel chosen so the judge tier is at or above the decision tier on the public 2026 Artificial Analysis index: Claude Opus 4.8 (within-family upgrade, strictly tier-above Opus 4.7), Google Gemini 2.5 Pro, and OpenAI GPT-5.5 (cross-family upgrades). The Gemini and GPT judges are reasoning-by-default models; we attach OpenRouter’s reasoning = {effort : minimal} control to both so the judge thinking budget matches the non-reasoning Anthropic baseline. Claude Haiku 4.5 from a pilot run is reported as a weak-judge reference. Every model is served through the OpenRouter unified API; observed cost is on the order of a few US dollars per full sweep of the 29-week test split.

E Full Per-Slice Results

This appendix expands the headline numbers of §5 into the per-slice audits underlying them.

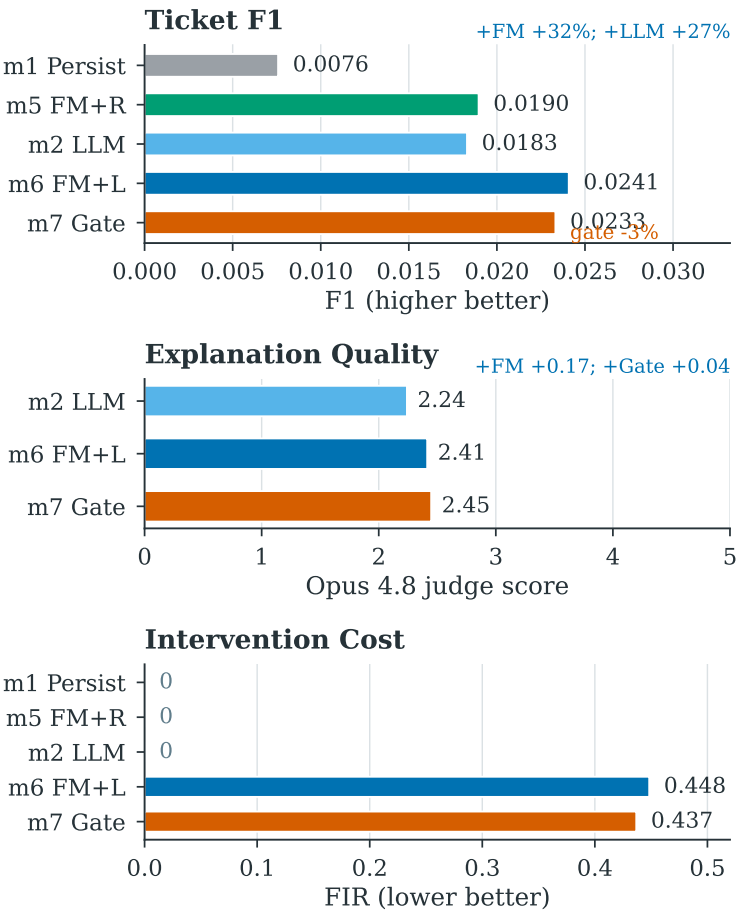


Figure 2: Layer-wise contribution and cost (visualising Tables 1 and 3). Adding the FM signal to a snapshot LLM improves ticket F1 and explanation quality; swapping the FM-fed rule engine for the LLM improves recall-oriented F1 further; the same LLM variants incur a substantial false-intervention rate, partially mitigated by the safety gate.

Table 4: b1_forecast all-horizon forecasting on the 2025 test split. PageRank MAE shown in 10^{-5} units. Best per horizon and metric in **bold**; ties share the bold. Trend consistency for m1_persistence is structurally 0 because $\hat{G}_{t+h} = G_t$ produces no trend signal.

h	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Trend Cons ($\uparrow$)
1	m1_persistence	3.6	0.0376	0.0027	0.556	0.000
	m3_evolvegc	3.7	0.0481	0.1149	0.535	0.444
	m4_fm_only	4.7	0.0376	0.0029	0.549	0.531
	m5_fm_rules	4.7	0.0376	0.0029	0.549	0.525
4	m1_persistence	3.4	0.0322	0.0018	0.570	0.000
	m3_evolvegc	3.5	0.0489	0.1326	0.551	0.324
	m4_fm_only	4.5	0.0321	0.0022	0.559	0.621
	m5_fm_rules	4.5	0.0321	0.0022	0.558	0.628
8	m1_persistence	3.9	0.0399	0.0028	0.517	0.000
	m3_evolvegc	4.0	0.0492	0.1140	0.499	0.296
	m4_fm_only	4.5	0.0398	0.0030	0.508	0.624
	m5_fm_rules	4.5	0.0398	0.0030	0.508	0.632
12	m1_persistence	3.7	0.0393	0.0027	0.525	0.000
	m3_evolvegc	3.8	0.0534	0.1245	0.509	0.314
	m4_fm_only	4.1	0.0392	0.0029	0.516	0.600
	m5_fm_rules	4.1	0.0392	0.0029	0.516	0.600

Table 5: b4_stress per-scenario detail. Loss MAE in normalised units; Distress count MAE in protocols; target overlap@10 measured against the actual future graph. Best per scenario and metric in **bold**; ties share the bold, and columns where all four methods tie are left unbolded. The main pattern is conservative: FM variants roughly match the strong persistence baseline on loss and target-overlap metrics, while EvolveGCN is generally weaker on downstream what-if fidelity.

Scenario	Method	Loss MAE ($\downarrow$)	Distress MAE ($\downarrow$)	Prop. Depth MAE ($\downarrow$)	Overlap@10 ($\uparrow$)
S1 (single protocol)	m1_persistence	0.0668	200.3	0.897	0.314
	m3_evolvegc	0.1072	218.1	0.966	0.034
	m4_fm_only	0.0667	201.4	0.897	0.314
	m5_fm_rules	0.0667	201.6	0.897	0.314
S2 (bridge cluster)	m1_persistence	0.0150	27.3	0.000	0.521
	m3_evolvegc	0.0116	196.4	0.000	0.189
	m4_fm_only	0.0151	50.3	0.000	0.525
	m5_fm_rules	0.0151	52.1	0.000	0.525
S3 (stablecoin de-peg)	m1_persistence	0.0079	0.0	0.000	0.445
	m3_evolvegc	0.0156	0.0	0.000	0.269
	m4_fm_only	0.0080	0.0	0.000	0.445
	m5_fm_rules	0.0080	0.0	0.000	0.445
S4 (sector lending)	m1_persistence	0.0100	0.0	0.034	0.536
	m3_evolvegc	0.0200	0.0	0.034	0.293
	m4_fm_only	0.0099	0.0	0.034	0.536
	m5_fm_rules	0.0099	0.0	0.034	0.536
S5 (correlated top-10)	m1_persistence	0.0100	0.0	0.000	0.602
	m3_evolvegc	0.0538	0.0	1.000	0.330
	m4_fm_only	0.0099	0.0	0.000	0.602
	m5_fm_rules	0.0099	0.0	0.000	0.602

Table 6: b6_robustness per-regime detail. All metrics at $h=4$ on the 2025 test split under five degradation regimes. PageRank, HHI, and Gini MAE in 10^{-3} units. Relative degradation is signed against the clean b1_forecast baseline; negative values mean the predictor is no worse than on clean data. Best per metric (lowest MAE, highest rank correlation, smallest $|\Delta_{\text{rel}}|$) in **bold**; ties share the bold. Persistence ignores node features, so its noisy_features_01 and missing_features_20 rows degenerate to the clean numbers (and $\Delta_{\text{rel}} = 0$); EvolveGCN’s predictions also change only negligibly under these two regimes ($|\Delta_{\text{rel}}| < 10^{-8}$).

Regime	Method	PR MAE ($\downarrow$)	HHI MAE ($\downarrow$)	Gini MAE ($\downarrow$)	Rank Corr ($\uparrow$)	Δ_{rel} (~ 0)
low_data_10pct	m1_persistence	15.94	0.0591	0.0023	0.552	+0.766
	m3_evolvegc	15.23	0.0999	0.1301	0.535	+ 0.276
	m4_fm_only	12.83	0.0586	0.0027	0.528	+0.612
	m5_fm_rules	12.71	0.0587	0.0027	0.528	+0.607
low_data_25pct	m1_persistence	9.46	0.0282	0.0020	0.577	−0.093
	m3_evolvegc	10.22	0.0486	0.1281	0.558	− 0.027
	m4_fm_only	9.86	0.0279	0.0024	0.553	−0.126
	m5_fm_rules	9.95	0.0279	0.0024	0.553	−0.126
partial_graph_30	m1_persistence	9.63	0.0350	0.0019	0.554	+0.068
	m3_evolvegc	10.36	0.0488	0.1236	0.537	− 0.048
	m4_fm_only	12.52	0.0350	0.0022	0.500	+0.082
	m5_fm_rules	12.45	0.0350	0.0022	0.499	+0.077
noisy_features_01	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegc	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	11.38	0.0323	0.0021	0.554	−0.005
	m5_fm_rules	11.35	0.0323	0.0021	0.554	−0.007
missing_features_20	m1_persistence	9.77	0.0322	0.0018	0.584	0.000
	m3_evolvegc	10.68	0.0489	0.1326	0.563	0.000
	m4_fm_only	12.11	0.0323	0.0023	0.559	+0.017
	m5_fm_rules	12.08	0.0323	0.0023	0.559	+0.014

Table 7: b2_warning per-event detail for the shared weighted-degree monitor (h1_weighted_degree) across alert budgets $K \in \{5, 10, 20\}$. Recall and F1-warning shown in 10^{-3} units (the full DeFi corpus contains $\sim 4,300$ protocols, so per-week recall denominators are large). Lead time is event-level and not budget-sensitive: 5 weeks for Terra/Luna and FTX, 4 weeks for SVB-USDC. b2_warning sits upstream of the predictor choice, so a single monitor is shared across all methods (see §5).

Event	K	Prec. ($\uparrow$)	Rec. $\times 10^3$ ($\uparrow$)	F1 _w $\times 10^3$ ($\uparrow$)	Stab. ($\uparrow$)	$ \mathcal{A} $
Terra/Luna	5	0.60	1.81	3.61	0.44	11
	10	0.70	3.26	6.49	0.62	19
	20	0.65	7.97	15.74	0.74	36
FTX	5	0.60	1.33	2.65	0.64	10
	10	0.70	2.39	4.77	0.72	18
	20	0.55	4.79	9.49	0.72	39
SVB-USDC	5	1.00	1.81	3.62	0.55	10
	10	1.00	2.82	5.62	0.75	15
	20	1.00	5.64	11.22	0.81	30

Table 8: Per-regime data-health gate ablation isolating A1 from A2. Each row holds $\tau_{\text{conf}}=0$ and sweeps τ_{data} . “strict” = $\tau_{\text{data}}=0.85$, “on” = 0.70, “off” = 0.00. The four regimes apply targeted edge or feature degradation on the clean 2025 test split ($n_{\text{weeks}}=29$): extreme masks both topology and features at 90%, topo removes 90% of edges only, feat corrupts 90% of node features only, severe masks both at 80%. Numbers expand Panel B of Table 9: under degradation the gate-off setting produces FIR up to 0.541, while *on/strict* can suppress intervention tickets via safe-mode and drive FIR to 0.

Regime	Gate	τ_{data}	$\overline{d\bar{h}}$	Safe-mode (%)	Ticket Prec. ($\uparrow$)	FIR ($\downarrow$)	Interventions
extreme	strict	0.85	0.587	100	0.510	0.000	0
	on	0.70	0.587	100	0.545	0.000	0
	off	0.00	0.587	0	0.490	0.510	57
extreme_topo	strict	0.85	0.637	100	0.517	0.000	0
	on	0.70	0.637	100	0.462	0.000	0
	off	0.00	0.637	0	0.559	0.441	58
extreme_feat	strict	0.85	0.755	100	0.503	0.000	0
	on	0.70	0.755	0	0.510	0.267	5
	off	0.00	0.755	0	0.503	0.267	5
severe	strict	0.85	0.751	100	0.490	0.000	0
	on	0.70	0.751	0	0.538	0.400	25
	off	0.00	0.751	0	0.538	0.541	24

E.1 Full all-horizon b1_forecast results

E.2 Per-scenario b4_stress results

E.3 Per-regime b6_robustness results

E.4 Per-budget b2_warning results

E.5 Per-regime A1 (data-health gate) isolation

E.6 Stress-condition ablation Panels B and C

Panel B: under severe feature/edge masking, FIR ranges over $[0.27, 0.60]$ when the data-health gate is disabled ($\tau_{\text{data}} = 0$; the confidence gate is held off at $\tau_{\text{conf}} = 0$ throughout Panel B to isolate A1), and collapses to 0.000 when the gate is triggered, by suppressing intervention tickets. Panel C: multi-horizon monitoring ($h \in \{1, 4, 8, 12\}$) generates 73–98 alerts in the three crisis windows vs 19–23 for single-horizon ($h = 4$), at unchanged precision range 0.533–0.624 and zero false interventions in both cases.

F Statistical Significance and Matched-Budget Analysis

All headline metrics are means over the $n = 29$ scored test weeks. This appendix reports week-level uncertainty for the central comparisons: 95% percentile bootstrap confidence intervals (10,000 resamples of weeks with replacement; F1 is recomputed from the resampled mean precision and recall) and two-sided paired sign-flip permutation tests (20,000 permutations, methods aligned by week). Analysis code is released with the harness (scripts/bootstrap_stats.py, scripts/matched_budget.py).

F.1 Confidence intervals

F.2 Paired permutation tests

Three readings follow. (i) The load-bearing quantitative claims—the FM signal’s F1 lift, the full stack’s F1 lift over rules, the Sonnet 4.6 F1 improvement, and the over-intervention cost—are significant at $p < 0.001$. (ii) The explanation-quality lift of the FM-grounded variants is judge-dependent: significant under GPT-5.5, positive but not significant under Opus 4.8 ($p = 0.23$), and absent under the (noisiest) Gemini judge. We therefore read the judge axis as directional evidence rather than a confirmed effect. (iii) The safety gate’s effects on F1 and judge score are indistinguishable from noise on clean data, consistent with its characterisation as a reserve mechanism (§6).

Table 9: Component-level ablation of Sentinel on b5_decision. Panel A shows clean 2025 test-split ablations against the m5_fm_rules baseline; Panels B–C report targeted stress runs for mechanisms that are dormant on clean data. A2 (confidence gating) and A3 (scenario engine) are load-bearing in the clean setting, while A1 (data-health gating) and A6 (multi-horizon monitoring) are safety-reserve mechanisms whose effects appear under data degradation and crisis-window dynamics, respectively.

Panel	Ablation	Configuration	Ticket Prec.	FIR	Notes
A: clean test	—	full m5_fm_rules	0.600	0.000	baseline
	A1	disable data-health gate ($\tau_{\text{data}}=0$)	0.600	0.000	no effect (clean data)
	A2	disable confidence gate ($\tau_{\text{conf}}=0$)	0.600	0.429	FIR shoots up
	A3	skip scenario engine	0.000	0.000	target extraction collapses
	A6	single-horizon forecasting ($h=4$)	0.600	0.000	no effect (clean data)
B: degraded data	A1 off	80–98% feature/edge mask, $\tau_{\text{conf}}=0$	—	0.27–0.60	false interventions emerge
	A1 on	same degradation, safe-mode triggered	—	0.000	intervention suppression
C: crisis period	A6: single	horizons = {4}, crisis windows	0.533–0.624	0.000	alerts: 19–23
	A6: multi	horizons = {1, 4, 8, 12}, same windows	0.533–0.624	0.000	alerts: 73–98 ($\approx 4\times$)

Table 10: 95% bootstrap CIs for the decision-quality headline numbers (decision LLM in parentheses; judge is Claude Opus 4.8 throughout). [‡]m1’s per-week distribution comes from a local re-run of the released harness: its aggregate F1 (0.0081) brackets the published GPU-run value (0.0076), and the recall@ k and stressed-pool size reproduce exactly, but with <1 ticket per week a few borderline tickets flip across environments (re-run precision 0.633 vs published 0.720), so the m1 interval should be read as order-of-magnitude.

Configuration	F1 [95% CI]	FIR [95% CI]	Judge [95% CI]
m1 rules [‡]	0.0081 [0.0041, 0.0123]	0.000	—
m2 (Opus 4.7)	0.0184 [0.0142, 0.0223]	0.000	2.24 [2.10, 2.41]
m6 (Opus 4.7)	0.0241 [0.0201, 0.0281]	0.448 [0.293, 0.601]	2.41 [2.21, 2.62]
m7 (Opus 4.7)	0.0234 [0.0194, 0.0271]	0.437 [0.282, 0.592]	2.45 [2.24, 2.66]
m6 (Sonnet 4.6)	0.0276 [0.0225, 0.0324]	0.000	2.52 [2.28, 2.79]
m7 (Sonnet 4.6)	0.0288 [0.0236, 0.0335]	0.000	2.66 [2.41, 2.90]
m6 (Gemini 2.5 Pro)	0.0129 [0.0103, 0.0155]	0.000	2.21
m7 (Gemini 2.5 Pro)	0.0139 [0.0111, 0.0167]	0.000	2.38

Table 11: Two-sided paired permutation p -values for the key comparisons. Significant effects ($p < 0.05$) in **bold**.

Comparison	Metric	p
m6 vs m2 (FM signal)	F1	0.0002
m7 vs m1 (full stack)	F1	<0.0001
m7 vs m6 (gate)	F1	0.41
m7 $\times$ Sonnet vs m7 $\times$ Opus	F1	0.0004
m6 vs m2	FIR	<0.0001
m6 vs m2 (Opus 4.8 judge)	Judge	0.23
m6 vs m2 (GPT-5.5 judge)	Judge	0.0002
m6 vs m2 (Gemini judge)	Judge	0.77
m7 vs m6 (Opus 4.8 judge)	Judge	1.00
m7 $\times$ Sonnet vs m7 $\times$ Opus	Judge	0.21

F.3 Matched-budget recall@ k

Methods emit different numbers of ticket targets per week (m1 2.1, m2 4.9, m7 6.3, m6 6.5, m7 $\times$ Sonnet 7.6), and unbudgeted recall mechanically favours methods that flag more protocols. To separate target *quality* from target *volume*, we truncate every method to its top- k targets per week (LLM variants ranked by their own emitted risk score; m1 by ticket confidence) and score against the same stressed pool.

The matched-budget view decomposes the headline F1 result into two distinct effects. Against the *rules baseline*, LLM targeting is genuinely better per target: at every k , precision@ k roughly doubles

Table 12: Matched-budget recall@ k ($\times 10^3$) and precision@ k on the 2025 test split. At matched budgets every LLM variant beats the rules baseline ($p \leq 0.0055$ vs m1 at all k), while the FM-grounded and raw-snapshot LLM variants are statistically indistinguishable from each other (all pairwise $p \geq 0.10$) and the top- k hit patterns of the three FM-fed decision LLMs nearly coincide. m1 row from the local re-run (Table 10 note).

Method	$k=1$		$k=2$		$k=3$	
	R@1	P@1	R@2	P@2	R@3	P@3
m1 rules	0.71	0.24	1.71	0.28	2.45	0.26
m2 (Opus 4.7)	1.53	0.52	3.47	0.55	5.70	0.59
m6 (Opus 4.7)	1.64	0.55	3.29	0.53	5.30	0.56
m7 (Opus 4.7)	1.64	0.55	3.47	0.55	5.30	0.56
m7 (Sonnet 4.6)	1.64	0.55	3.47	0.55	5.30	0.56

(0.52–0.59 vs 0.24–0.28) and recall@ k is significantly higher ($p \leq 0.0055$). Against the *raw-snapshot LLM*, however, the FM’s contribution is not a higher per-target hit rate at the top of the ranking—at matched k the FM-fed and raw-snapshot variants are statistically indistinguishable, and the FM-fed variants’ top- k targets are nearly identical across decision LLMs because the evidence bundle pins the highest-exposure protocols. The FM’s F1 gain comes from supporting a *larger* target set at undiminished precision (Table 1), which is also where the over-intervention risk concentrates: under Opus 4.7 the expanded tail is escalated to intervention-level severity (FIR = 0.448), while Sonnet 4.6 expands the same set without escalating it (FIR = 0).